Dynamic Planet C - Dynamic Planet C - University of Chicago C - 01-22-2022

Hello! This is Dynamic Planet. You'll have 50 minutes to complete the test.

Please read the following instructions carefully:

This test consists of a variety of different questions. If your answer is not clear, it will get marked as incorrect. If a question is worth any points, you'll be able to see. Ties will be broken automatically using a few pre-designated questions from this exam. You are strictly prohibited from using the internet. You are only allowed to use resources prepared beforehand in accordance with the event rules - though those may be accessed online. Please make this a fair test for everyone! Don't be that person who cheats :((You may only resort to your notes and your partner. You won't lose points for wrong/unanswered questions so I highly recommend dividing the workload and trying both of your best!

Good luck and have fun! :)

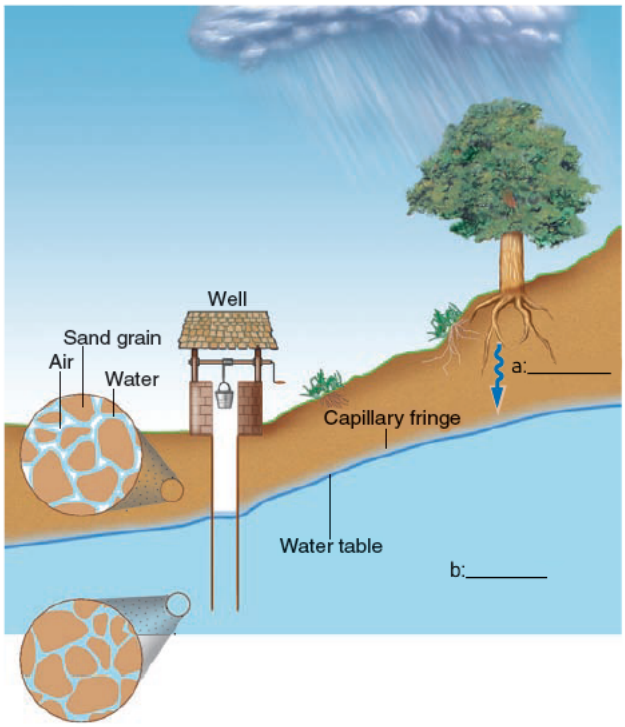
1. (1.00 pts) Define groundwater.

Expected Answer: Groundwater is water that occurs below the ground surface in zones of porous materials where all the cracks (fractures) and pore spaces are filled with water and that water is at a pressure higher than the pressure of the atmosphere.

2. (1.00 pts) How much of the hydrosphere is groundwater?

Expected Answer: ~1.7%

3. (2.00 pts) Fill in the blanks



Expected Answer: zone of aeration, unsaturated zone, or vadose zone////zone of saturation, saturated zone, or groundwater zone

4. (1.00 pts) There is ___ atmospheric pressure above the water table and ___ atmospheric pressure below the water table.

- A) positive, negative
- B) negative, negative
- C) negative, positive
- D) positive, positive

5. (1.00 pts) Natural groundwater is commonly:

- A) very acidic
- B) very basic
- C) slightly acidic
- D) slightly basic
- E) pretty much neutral

6. (1.00 pts) Explain your reasoning for the previous question.

Expected Answer: Natural groundwater is commonly slightly acidic because of dissolved carbon dioxide (CO_2) from the atmosphere or from soil gases

7. (2.00 pts) Describe the role that groundwater plays in karst formations. What chemically allows it to occur?

Expected Answer: Most caves develop when slightly acidic groundwater dissolves limestone along joints and bedding planes, opening up cavern systems as calcite is carried away in solution. In areas where groundwater dissolves carbonate rock, a distinctive landscape may develop that is called karst topography.

8. (2.00 pts) Would you expect to find a cavern in granite? Would you expect to discover a cavern in shale? Defend your answers.

Expected Answer: Yes in granite, it can dissolve in water. Not in shale, it is hard to dissolve.

9. (2.00 pts) The drop in the water table around a pumped well is the

- (●) A) drawdown
- ( ) B) hydraulic head
- ( ) C) porosity
- ( ) D) fluid potential

10. (3.00 pts) Define the terms in Darcy's law: v, Q, K, Δh, ΔL, A

Expected Answer: v = velocity of groundwater flow (m/s) Q=volumetric flow rate/groundwater discharge (m3/s) (optional) K=saturated hydraulic conductivity, is the proportionality constant reflecting the ease with which water flows through a material (m/s) Δh=difference in hydraulic head between two measuring points (m) ΔL=length along the flow path between locations where hydraulic heads are measured (m) A=cross-sectional area of flow perpendicular to the direction of flow (m2) (optional)

11. (4.00 pts) Describe four different types of topographical drainage networks. What factors are responsible for the formation of each?

Expected Answer: Radial: drainage network on cone-shaped mountain, looks like spokes on wheel Rectangular: occur where a rectangular grid of fractures breaks the ground, look rectangular Trellis: drainage network develops across a landscape of parallel valleys and ridges, major tributaries flow down a valley and join a trunk stream that cuts across the ridges, looks like a garden trellis Parallel: on a uniform, fairly steep slope, several streams with parallel courses develop simultaneously. Streams form a parallel network

12. (1.00 pts) Why are there variations in velocity in the same stream?

- A) the gravitational pull of the earth is stronger at some points along the stream
- B) stochastic variation
- C) wind at different points along the stream cause its speed to change
- D) friction along the walls and bed of the stream slows the flow of water

13. (2.00 pts) Name at least 2 factors that control a stream's velocity.

Expected Answer: gradient, channel shape, discharge

14. (1.00 pts) The broad strip of land built up by sedimentation on either side of a stream channel is

- A) a flood plain
- B) a delta
- C) an alluvial fan
- D) a meander

15. (1.00 pts) A river's velocity is _____ on the outside of a meander curve compared to the inside.

- A) equal
- B) lower
- C) higher

16. (1.00 pts) A stream that doesn't meander at all has a sinuosity of___.

1

17. (3.00 pts) How do clastic sediments differ from organic sediment and chemical sediment?

Expected Answer: Clastic sediment is sediment composed of fragments of weathered rock called clasts. Clastic sediment is rounded and sorted during transport and then depositec Chemical sedimentary rocks are rocks which are formed due to chemical precipitation or any other chemical processes like evaporation. Sedimentary rocks which are formed due to biogenic activity/processes or form from organic matter are called organic sedimentary rocks.

18. (1.00 pts) The competence of a stream is

- A) its speed
- B) the maximum particle size it carries
- C) the total quantity of sediment it can carry
- D) the amount of water in the stream

19. (1.00 pts) The capacity of a stream is

- A) its speed
- B) the maximum particle size it carries
- C) the total quantity of sediment it can carry
- D) the amount of water in the stream

20. (1.00 pts) Coarser sediment tends to settle out _____ and finer sediment settles out _____.

- A) upstream, downstream
- B) downstream, upstream
- C) north of the equator, south of the equator
- D) south of the equator, north of the equator

21. (3.00 pts) How does acidification affect algal populations?

Expected Answer: It causes them to increase and then collapse, which depletes O2 in the water.

22. (2.00 pts) Select all that would help counter the effects of acid precipitation.

(Mark **ALL** correct answers)

- [x] A) burning low-sulfur fuels for industry and power production and decreasing the emission of nitrogen compounds in automobile exhaust
- [ ] B) adding lye (sodium hydroxide) to lakes and watersheds to neutralize the effects of the acid
- [x] C) adding lime (calcium carbonate) to lakes and watersheds to neutralize the effects of the acid
- [ ] D) adding hydrochloric acid to lakes and watersheds to neutralize the effects of the acid
- [ ] E) increasing methane emissions

23. (1.00 pts) Which of the following pollution sources is geogenic?

A) Agricultural fertilizer runoff
B) Leakage from septic tanks into aquifers
C) Radioactive leakage from bedrock into aquifers
D) Microplastic in rivers

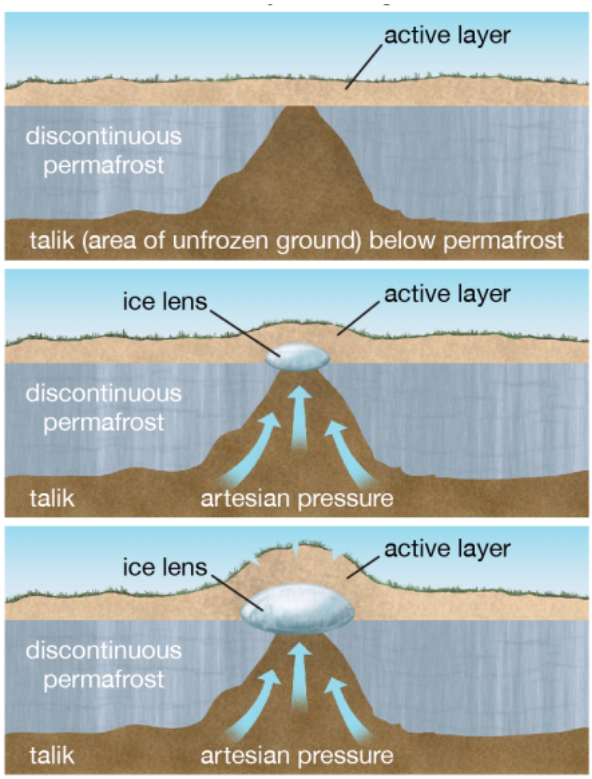
24. (1.00 pts)
Aquifers provide water for millions of people. But when aquifers are used at faster rate than they can replenished a deficit occurs which causes a number of problems.

What is it called when aquifers in coastal areas start to fill with ocean water?

A) Confined
B) Saltwater intrusion
C) Artesian
D) Depletion
E) Groundwater

25. (1.00 pts)

The following diagram shows the formation process for what feature?



A) Tundra well
B) Glacial mound
C) Artesian well
D) Open-system pingo
E) Karst window
F) Closed-system pingo

26. (1.00 pts) What agent of erosion is mostly responsible for the depressions occupied by both the kettle lakes and finger lakes found in New York State?

- A) Wind
- B) Glaciers
- C) Waves
- D) Streams

27. (2.00 pts) What causes a turbidity current?

Expected Answer: A turbidity current is when water quickly goes downhill because of increased density. The increased density in a turbidity current is caused by a lot of sediment. More sediment piles up when mud and sand is loosened by events like earthquakes and landslides.

28. (1.00 pts) The instrument used for measuring evaporation is called a:

- A) Hygrometer
- B) Turbidometer
- C) Atmometer
- D) Lysimeter
- E) Ombrometer
- F) Phytometer

29. (1.00 pts) What defines a rainfall intensity of light rain?

- A) When the rainfall period lasts for under 10 minutes.
- B) When the precipitation rate is between 2.5 to 7.6 mm per hour.
- C) When the rain droplets are less than 1 cm in length.
- D) When the precipitation rate is <2.5 mm per hour.

30. (1.00 pts) Which of the following factors affect water infiltration of a formation?

(Mark **ALL** correct answers)

- [x] A) Type of soil medium
- [x] B) Vegetative cover
- [x] C) Soil compaction
- [ ] D) Wind velocity
- [ ] E) Soil acidity
- [ ] F)

31. (1.00 pts) A hydrograph is the graphical representation of:

- A) runoff and time
- B) rainfall and time

C) groundwater flow and time
D) porosity and grain size
E) discharge and size

32. (1.00 pts) Which of the following are point-source pollution?

(Mark **ALL** correct answers)
- [x] A) Sewage treatment plant
- [x] B) Paper mill
- [ ] C) Automobile emissions
- [ ] D) Runoff from a farm
- [ ] E) Runoff from a city

33. (1.00 pts) All of the following characteristics apply to the riparian zone EXCEPT:

- A) high species diversity
- (•) B) high sun exposure
- C) wildlife corridor
- D) biofilters

34. (1.00 pts) The uppermost level at which the water in a given area fully saturates rock or soil is called a:

- A) Impermeable surface
- B) Groundwater recharge
- (•) C) Water table
- D) Mantle level

35. (1.00 pts) What are other names for disappearing streams?

- A) Karst stream
- (•) B) Losing stream
- C) Vanishing stream
- D) Ephemeral spring

36. (1.00 pts) Which of the following are true about bogs?

(Mark **ALL** correct answers)
- [ ] A) Have an impermeable clay layer
- [x] B) Low in nutrients and highly acidic
- [ ] C) High in nutrients and highly acidic
- [x] D) Located in cold, temperate climates
- [ ] E) Located in tropical climates
- [x] F) Type of wetland that accumulates peat

37. (3.00 pts) If an area is susceptible to erosion, what is a land-use change that could be implemented to prevent further erosion?

Expected Answer: the incorporation of soil conservation measures (planting cover crops on farmland and establishing grassed waterways) can significantly reduce erosion and stream sediment load

38. (3.00 pts) What are negative environmental impacts of man-made dams?

Expected Answer: Dams alter aquatic ecology and river hydrology upstream and downstream, affecting water quality, quantity and breeding grounds. Can trap sediment, destroy habitats, cause flooding

39. (1.00 pts) What is the deepest layer in a thermally stratified lake called?

- A) Thermocline
- B) Epilimnion
- C) Hypolimnion
- D) Metalimnion
- E) Bathylimnion

40. (2.00 pts) Explain how and why a lake turnover occurs.

Expected Answer: Due to seasonal changes in sunlight intensity, surface water temperature begins to transition in the spring and fall. In the spring, the water surface warms. This causes the temperature of the top and bottom layers of the lake to equalize. With the help of strong winds, this new equilibrium breaks the thermal stratification, and the lake is able to mix. Oxygen from the surface mixes with the bottom, while nutrients trapped near the bottom are free to mix throughout the lake.

41. (2.00 pts) What is the difference between the Strahler and Shreve methods for determining stream order?

Expected Answer: Shreve: Unlike the Strahler method, at a confluence of two streams the two numbers are added together. Whereas for Strahler, if two streams of the same order merge, the resulting stream is given a number that is one higher. If two rivers with different stream orders merge, the resulting stream is given the higher of the two numbers.

42. (1.00 pts) What does it mean to delineate a watershed on a topographic map?

Expected Answer: Drawing contour lines on a topo map to identify a watershed's boundaries

43. (2.00 pts) How is a natural levee formed?

Expected Answer: A stream typically reaches its greatest velocity when it is close to flooding over its banks. This is known as the bank-full stage. As soon as the flooding stream overtops its banks and occupies the wide area of its flood plain, the water has a much larger area to flow through and the velocity drops significantly. At this point, sediment that was being carried by the high-velocity water is deposited near the edge of the channel, forming a natural bank or levée.

44. (2.00 pts) What is the difference in suspension caused by laminar flow vs. turbulent flow?

Expected Answer: streams that flow fast tend to be turbulent (flow paths are chaotic and the water surface appears rough) and the water may be muddy, while those that flow more slowly tend to have laminar flow (straight-line flow and a smooth water surface) and clear water. Turbulent flow is more effective than laminar flow at keeping sediments in suspensior

45. (1.00 pts) According to the Davis cycle of erosion, what characterizes an old-age stream?

(Mark **ALL** correct answers)

- [ ] A) Steep gradients of surrounding terrain
- [x] B) Meandering
- [x] C) Slow movement
- [x] D) Wide floodplain
- [ ] E) High erosion
- [ ] F) V-shaped valleys

46. (1.00 pts) How is an oxbow lake formed?

(Mark **ALL** correct answers)

- [ ] A) Tectonic activity
- [x] B) Erosion
- [x] C) Deposition
- [ ] D) Increased stream velocity
- [ ] E) Flooding

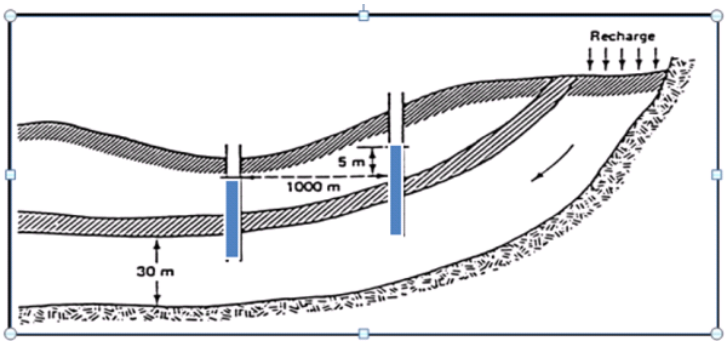
47. (4.00 pts) Solve for the 1) value of Q AND 2) the value of V (Darcy velocity) using Darcy's Law given the following information:

1. A confined aquifer has a source of recharge.

2. K for the aquifer is 50 m/day, and porosity n is 0.2.

3. The piezometric head in two wells 1000 m apart is 55 m and 50 m respectively, from a common datum.

4. The average thickness of the aquifer is 30 m, and the average width of the aquifer is 5 km = 5000m.



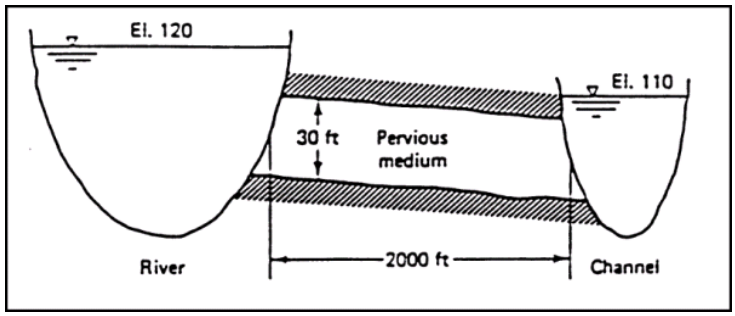
Expected Answer: Cross-Sectional area= 30(5000) = 1.5 x 10E5 m2 Hydraulic gradient dh/dL= (55-50)/1000 = 5 x 10E-3 Find Rate of Flow for K = 50 m/day Q = (50 m/day) (1.5 x 10E5 m2) (5 x 10E-3) Q = 37,500 m^3/day //////// Darcy Velocity: V = Q/A = (37,500m^3/day) / (1.5 x 10E5 m2) = 0.25m/day

48. (4.00 pts) Determine the flow rate Q of the seepage from the river to the channel given the following information:

1. A channel runs almost parallel to a river, and they are 2000 ft apart.

2. The water level in the river is at an elevation of 120 ft . The channel is at an elevation of 110ft.

3. A previous formation averaging 30 ft thick and with hydraulic conductivity K of 0.25 ft/hr joins them.



Expected Answer: Consider 1-ft (i.e. unit) lengths of the river and small channel. Where: A = (30 x 1) = 30 ft2 K = (0.25 ft/hr) (24 hr/day) = 6 ft/day Therefore, Q = [6ft/day (30ft2) (120 – 110ft)] / 2000ft///// Q = 0.9 ft3/day for each 1-foot length

49. (1.00 pts) Which of the following would describe a glacial lake?

- A) Mesotrophic
- B) Oligotrophic
- C) Eutrophic
- D) Hypotrophic

E) Hypertrophic

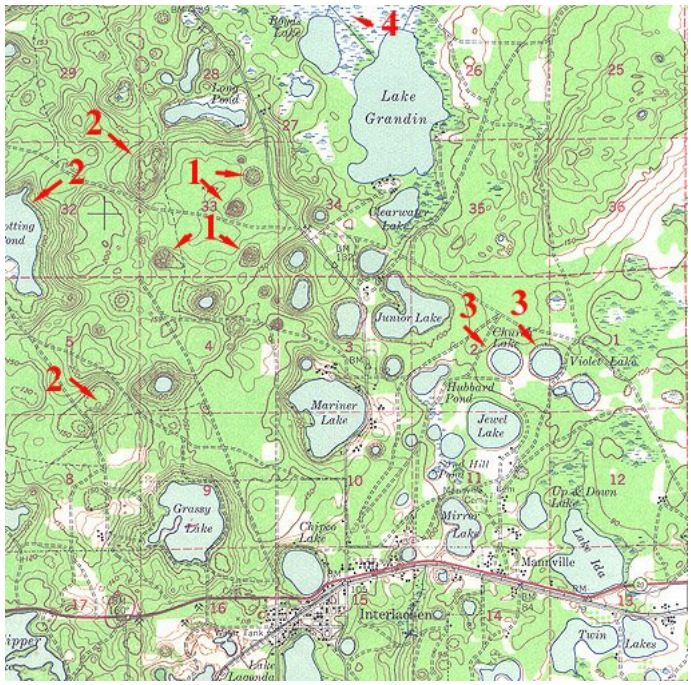
50. (1.00 pts) Describe 2 ways that nick points (or knickpoints) are formed.

Expected Answer: Uplift along a fault, glaciation resulting in a hanging valley, varying lithology including shale among igneous rocks, steeply sloped rivers

51. (3.00 pts) Explain the components of Chezy's equation, and what the equation is used to calculate.

Expected Answer: For a steady uniform flow of liquid through an open channel, the rate of flow of fluid or the discharge is calculated with Chezy's equation. v is average velocity, m or r is the hydraulic mean depth or hydraulic radius, 'i' is the slope of the bed and 'C' is the Chezy's Constant.

Use the map of a karst topography and identify the following features in the next 4 questions.



52. (1.00 pts) What feature is shown by arrow 1?

sinkhole | sinkholes

53. (1.00 pts) What feature is shown by arrow 2?

solution valleys | basin | solution valley

54. (1.00 pts) What feature is shown by arrow 3?

solution lakes | solution lake

55. (1.00 pts) Arrow #4 shows a large solution lake that has been invaded by vegetation. What has formed?

swamp | bog

Congratulations for finishing the test and best of luck for your other events!

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